



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of seven printed pages.*

SECTION A

(Attempt all questions from this Section)

Question 1.

Choose the correct answer to the questions from the given options. (Do not copy the Question, write the correct answer only):

[15]

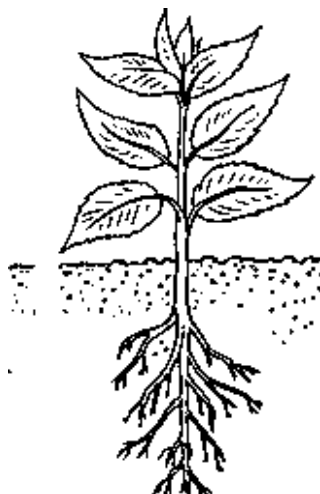
- (i) Osmosis and diffusion are the same except that in osmosis there is:
(a) Freely permeable membrane
(b) A cell wall in between
(c) A selectively permeable membrane in between
(d) Requires ATP
- (ii) In cohesion-tension theory, cohesion refers to the:
(a) Attraction between water molecules
(b) Attraction between the walls of xylem vessel
(c) Attraction between water molecules and xylem
(d) Attraction between water molecules and phloem.
- (iii) Riddhi tried to understand the function of two blood vessels in human body. She tabulated the pairs as follows:

Blood vessel	Function
P	Carries unfiltered blood
Q	Carries filtered blood

- (a) P- Renal Vein and Q- Renal Artery
(b) P- Posterior Vena cava and Q- Renal Artery
(c) P- Renal Vein and Q- Dorsal Aorta
(d) P- Renal Artery and Q- Renal Vein

- (iv) Shyam was skating on the road. He observed an alcoholic person who was walking clumsily. He was not being able to balance himself or even stand properly. This is due to:
- Improper functioning of middle ear
 - Improper functioning of cerebellum
 - Improper functioning of cerebrum
 - Improper functioning of medulla oblongata
- (v) Assertion (A): Malpighian capsule is present in the cortex of the kidney.
Reason (R): Bowman's Capsule and Proximal Convoluted Tubule are present in the cortex and gives dotted appearance.
- A is true and R is false
 - Both A and R are true
 - A is false and R is true
 - Both A and R are false
- (vi) Akshita was studying about the ovulatory phase of a woman. She was discussing this with her friends and they came to a conclusion that few hormones are involved in this phase. Identify who had given the correct answer:
- Rina- Progesterone and FSH
 - Akshita- LH and Progesterone
 - Sohini- LH and FSH
 - Alam- Oestrogen and LH
- Rina is correct
 - Alam is correct
 - Sohini is correct
 - Akshita is correct
- (vii) If a pure inflated pod plant (CC) is crossed with a pure constricted pod plant (cc), the phenotypic ratio (P) and genotypic ratio (Q) of F₂ generation will be:
- P- All inflated pod and Q all constricted pod
 - P- 3 inflated: 1 constricted and Q- 1CC: 2Cc : 1cc
 - P- 1 constricted: 2 inflated and Q- 3CC: 1cc
 - P- 1 inflated: 3 constricted and Q- 2CC: 1cc: 1Cc
- Only 1
 - Only 2
 - Only 3
 - Only 4
- (viii) Among the following which nitrogenous base is not present in DNA?
P. Uracil Q. Adenine R. Thymine S. Cytosine
- Only P
 - Only R
 - Both Q and S
 - Only Q
- (ix) Assertion (A) - During photosynthesis Carbon dioxide is reduced to form glucose.
Reason (R) – Carbon dioxide loses electron so it is reduced.
- A is true and R is false
 - Both A and R are true
 - A is false and R is true
 - Both A and R are false
- (x) Sayani in her class was trying to observe the board from the last row and read the instructions written by her teacher. She found out that she could not read the words clearly. Identify the problem which made Sayani not read the instructions written on the board from a distance.
- Cataract
 - Hyperopia
 - Myopia
 - Squint

- (xi) ____P____ prepares the body for action against abnormal conditions and ____Q____ is concerned with re-establishing normal condition in the body after an action. Identify P and Q:
- P- Central Nervous System and Q- Adrenal medulla
 - P- Somatic Nervous System and Q- Automatic Nervous System
 - P- Parasympathetic Nervous System and Q- Sympathetic Nervous System
 - P- Sympathetic Nervous System and Q- Parasympathetic Nervous System
- (xii) Hormone which helps in the contraction of muscles of uterine walls in females:
- TSH
 - Glucagon
 - Oxytocin
 - FSH
- (xiii) The walls of the left ventricle are thicker than the walls of the other chambers. Which of the following reasons support this statements?
- P- It pumps blood to different parts of the body.
Q- It pumps blood to the lungs.
- Only P
 - Both P and Q
 - Only Q
 - Neither P nor Q
- (xiv) Below is the diagram of a plant growing in the garden. If too much of DDT is added to the plant we will see the plant will wilt due to the changes in the external condition. What will be the process which caused the plant to wilt? Identify the tonicity of the solution outside the plant.



- Endosmosis and Hypertonic
 - Exosmosis and Hypotonic
 - Endosmosis and Hypotonic
 - Exosmosis and Hypertonic
- (xv) Assertion (A) – Mineral required for the functioning of thyroid gland is Iodine.
Reason (R) – Use of Iodized salt is recommended for proper functioning of Thyroid gland.
- A is true and R is false
 - A is false and R is true
 - Both A and R is true
 - Both A and R are false.

Question 2

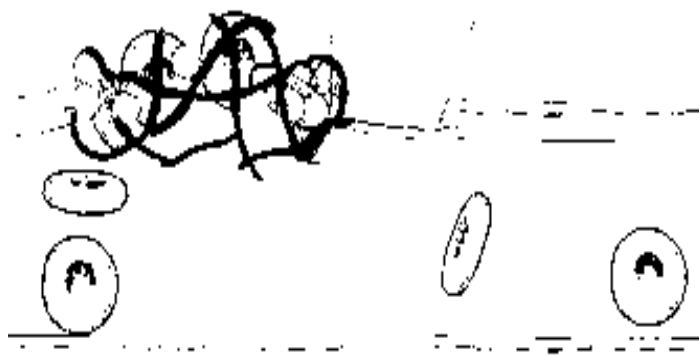
(i) Name the following:

[5]

- (a) Enzyme present in tear.
- (b) A plant with Trichomes.
- (c) A condition in which glucose is present in the urine.
- (d) Protein which helps in coiling and packaging of DNA into nucleosomes.
- (e) The part of the chloroplast where light dependent reaction of photosynthesis takes place.

(ii) Given below is the diagram of blood clotting process. Read the information given below the diagram and fill in the blanks:

[5]



The ____ (a) ____ acts as an enzyme and with the help of ____ (b) ____ ion present in the plasma, it converts a substance prothrombin of the plasma, into ____ (c) ____.

Vitamin ____ (d) ____ is essential for the production of prothrombin. ____ (e) ____ is the soluble protein present in blood plasma.

(iii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined.

[5]

- (a) Cornea, Lens, Vitreous humour, Conjunctiva.
- (b) Luteal phase, Menstrual phase, Ovulatory phase, Follicular phase.
- (c) Sperm duct, Sperm, Vagina, Urethra
- (d) Insect, Frog, Grass, Snake
- (e) Nucleosome, Chromatin fibre, DNA, Nucleus

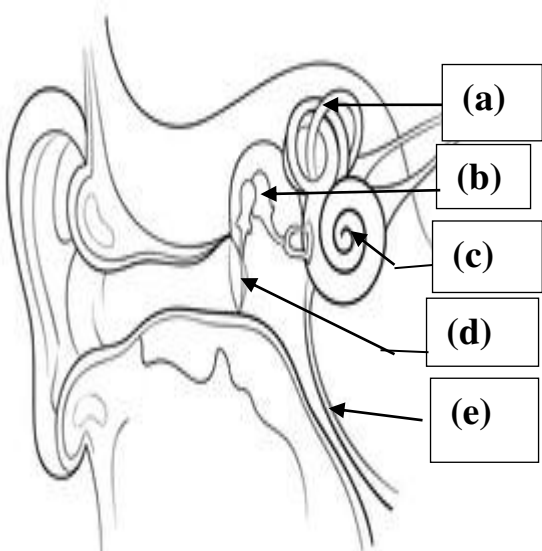
(iv) Read the explanation given below and name the structure:

[5]

- (a) Layer in plant cell which has hydrophilic substances in it.
- (b) Layer in the eye which has the pigment melanin in it.
- (c) Gland which secretes mineralocorticoids.
- (d) One half arm of a chromosome.
- (e) Blood vessel which carries maximum amount of glucose after digestion.

- (v) Given below is the cross section of human ear. Match the structures marked (a) To (e) with their correct functions:

[5]

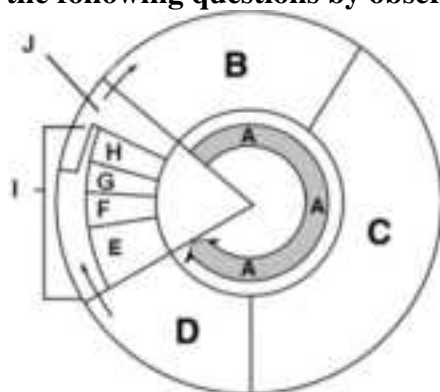
Cross section of Human Ear	Functions
	- Helps in hearing - Equalizes air pressure on both the sides of the eardrum. - Vibrates and converts sound waves to vibrations. - Maintains dynamic balance. - Vibrates and pass on the vibrations to the next ear ossicle. - Carries the sound waves towards the tympanum

SECTION B

(Attempt any four questions from this section)

Question 3

- Mention the name of a non-circulating fluid present in the front chamber of human eye. [1]
- Marine fish burst when kept in tap water. Give reasons. [2]
- A heterozygous tongue roller female and a homozygous Non Tongue Roller male produces off springs. Find out the percentage of offsprings who are Heterozygous Tongue Roller and percentage of offsprings who are Non Tongue Rollers. [2]
- ACTH is known as a tropic hormone. Explain the statement. [2]
- Answer the following questions by observing the diagram: [3]



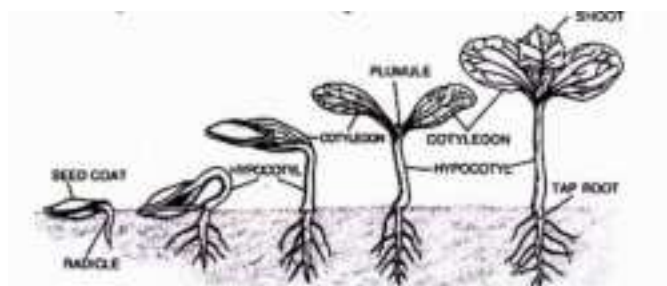
- Identify C and J.
- Mention two properties of F

Question 4

- (i) Define Photosynthesis. [1]
- (ii) Mention the exact location of SAN and Cuticle. [2]
- (iii) Differentiate between Embryo and Foetus. [2]
- (iv) Why do males suffer from colour blindness more than the females? [2]
- (v) Draw a neat labelled diagram of L.S. of Malpighian Capsule. [3]

Question 5

- (i) Haemophilia is a genetic disorder in which the sufferers are at risk of bleeding to death. What are the chances of a girl child to be haemophilic if her mother is normal and father is haemophilic? [1]
- (ii) Why do we boil a leaf in water during starch test? Name a plant with variegated leaves? [2]
- (iii) Explain the Law of Segregation. [2]
- (iv) Mention any two characteristics of root to absorb water from the soil. [2]
- (v) Study the diagram given below and answer the following questions: [3]



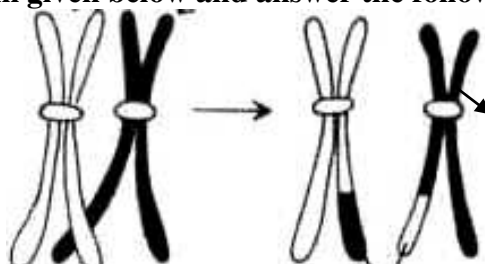
- (a) Account for the tropic movement of the plumule and radicle in response to light in the above diagram.
- (b) Define Tropic movement.
- (c) Seed kept in the soil show germination due to which two processes?

Question 6

- (i) Define "Bleeding" in plants. [1]
- (ii) Mention the exact location of pituitary gland. Mention the hormones secreted by Posterior lobe of pituitary gland. [2]
- (iii) Medha was observing few animal cells under the microscope which was undergoing mitotic division. She observed a cell where the nuclear membrane disappeared and centrioles were moving towards the opposite poles. Identify this stage and mention the name of the next stage which comes after this stage. [2]
- (iv) Draw a neat labelled diagram of the phase which was being observed by Medha as mentioned in question (iii) above. [2]
- (v) Pure Tall plant with Green pod (TTGG) is crossed with pure Dwarf plant with Yellow pod (ttgg). [3]
 - (a) Mention the phenotype of F₁ generation.
 - (b) Mention the probability of gametes that are formed in the F₂ generation.
 - (c) Mention the phenotypic ratio of F₂ generation.

Question 7

- (i) Mention the name of the paper which is used as an indicator of moisture. [1]
- (ii) Study the diagram given below and answer the following questions: [2]

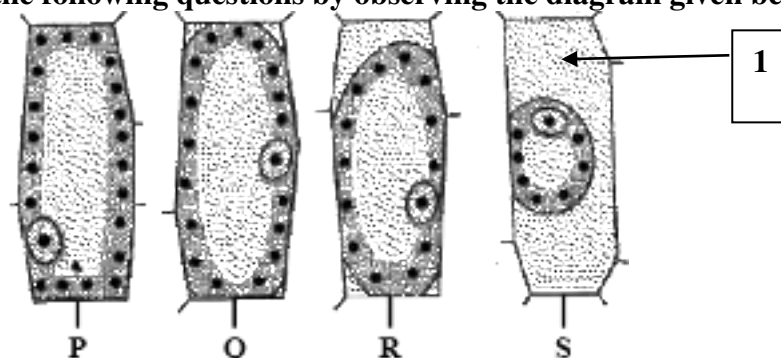


Identify the process and mention one significance of this process.

- (iii) Blood group “O” is known as Universal Donor. Explain with reasons. [2]
- (iv) When we have throat infection we gargle with saline water. Explain why. [2]
- (v) Draw a neat labelled diagram of root hair cell in turgid condition. [3]

Question 8

- (i) Mention the effect of high humidity on the rate of transpiration. [1]
- (ii) Answer the following questions by observing the diagram given below: [2]



- (a) Identify ‘1’ in the above diagram.
- (b) Mention the tonicity of the solution in which the cell ‘P’ was placed and how will we bring the cell “S” back to its normal state again?
- (iii) Name the pores present in the margin of the leaves which helps in guttation. Why does the portion below the portion of the stem from where the phloem is removed in girdling experiment shrinks? [2]
- (iv) Mention the name of the four forces which contribute to Ascent of Sap. [2]
- (v) Rima was conducting an experiment to show carbon dioxide is necessary for Photosynthesis. Answer the following questions in relation to the experiment: [3]
 - (a) Name the compound which absorbs carbon dioxide.
 - (b) What will be the observation on the experimental leaf and the control leaf with Iodine solution?